

Practice 1

7种音符，三个八度（低一、中、高一），三种升降调（降、原、升）

```
frequencies = zeros(7, 3, 3);

for tone = 1:7
    for nOctave = -1:1
        for rising = -1:1
            frequencies(tone, nOctave+2, rising+2) = tone2freq(tone, nOctave,
rising);
        end
    end
end

frequencies
```

```
frequencies =
frequencies(:,:,1) =

1.0e+03 *

    0.2077    0.4153    0.8306
    0.2331    0.4662    0.9323
    0.2616    0.5233    1.0465
    0.2772    0.5544    1.1087
    0.3111    0.6223    1.2445
    0.3492    0.6985    1.3969
    0.3920    0.7840    1.5680
```

```
frequencies(:,:,2) =

1.0e+03 *

    0.2200    0.4400    0.8800
    0.2469    0.4939    0.9878
    0.2772    0.5544    1.1087
    0.2937    0.5873    1.1747
    0.3296    0.6593    1.3185
    0.3700    0.7400    1.4800
    0.4153    0.8306    1.6612
```

```
frequencies(:,:,3) =

1.0e+03 *

    0.2331    0.4662    0.9323
    0.2616    0.5233    1.0465
    0.2937    0.5873    1.1747
    0.3111    0.6223    1.2445
    0.3492    0.6985    1.3969
    0.3920    0.7840    1.5680
    0.4400    0.8800    1.7600
```

Practice 2

3种升降调, 7种调号

```
% tone2freq(tone, scale, noctave, rising)
frequencies = zeros(3, 7);

for rising = -1:1
    for scale = 1:7
        frequencies(rising + 2, scale) = tone2freq(1, scale, 0, rising);
    end
end

frequencies
```

```
frequencies = 3x7
    246.8231    277.0271    311.0066    329.4121    369.5268    415.3047    466.2739
    261.5000    293.5000    329.5000    349.0000    391.5000    440.0000    494.0000
    277.0496    310.9524    349.0931    369.7526    414.7798    466.1638    523.3748
```

Practice 3

Test

```
fs = 8192;
melody = [gen_wave_3(1,1,0,0,1,fs), gen_wave_3(2,1,0,0,1,fs),
gen_wave_3(3,1,0,0,1,fs)]
```

```
melody = 1x24576
         0         0.1992         0.3905         0.5661         0.7190         0.8431         0.9333         0.9861
    0.9994    0.9726    0.9068    0.8046    0.6702    0.5089    0.3272    0.1323
   -0.0678   -0.2653   -0.4521   -0.6207   -0.7645   -0.8776   -0.9555   -0.9951
   -0.9948   -0.9546   -0.8761   -0.7625   -0.6183   -0.4493   -0.2623   -0.0648
    0.1354    0.3301    0.5115    0.6725    0.8065    0.9081    0.9733    0.9995
    0.9856    0.9322    0.8414    0.7169    0.5636    0.3877    0.1962   -0.0031
   -0.2023   -0.3933
```

```
sound(melody, fs)
```

天空之城

1=D $\frac{4}{4}$

$\dot{1}$ $\underline{6\ 7}$ $\parallel$ $\dot{1}$ $\underline{7}$ $\dot{1}$ $\dot{3}$ | 7 - - $\underline{3\ 3}$ | $\overset{\frown}{6\cdot}$ $\underline{5}$ 6 $\dot{1}$ | 5 - 0 $\underline{3\ 3}$ | 4 $\cdot$ $\underline{3}$ 4 $\dot{1}$
 | 3 - 0 $\underline{\dot{1}\ \dot{1}\ \dot{1}}$ | 7 $\cdot$ $\underline{4}$ $\underline{4}$ 7 | 7 - 0 $\underline{6\ 7}$ | $\dot{1}$ $\underline{7}$ $\dot{1}$ $\dot{3}$ | 7 - 0 $\underline{3\ 3}$ | 6 $\cdot$ $\underline{5}$ 6 $\dot{1}$
 | 5 - - 0 $\underline{3}$ | 4 $\underline{\dot{1}\ 7}$ $\overset{\frown}{7\ \dot{1}}$ | $\underline{\dot{2}\ \dot{2}\ \dot{3}}$ $\dot{1}$ 0 | $\dot{1}\ 7$ $\underline{6\ 6}$ 7 $\underline{5^\#}$ $\parallel$ 6 - 0 $\underline{\dot{1}\ \dot{2}}$ | 3 $\cdot$ $\underline{2}$ 3 5
 | $\dot{2}$ - 0 $\underline{5\ 5}$ | $\underline{\dot{1}\ 7}$ $\dot{1}$ $\dot{3}$ | 3 - 0 0 | $\underline{6\ 7}$ $\dot{1}\ 7$ $\underline{\dot{2}\ \dot{2}}$ | $\dot{1}$ $\overset{\frown}{5}$ 5 0 | 4 3 $\dot{2}$ $\dot{1}$
 | $\dot{3}$ - - - | 3 - 0 3 | 6 - 5 5 | $\underline{3\ 2}$ $\dot{1}$ 0 $\underline{\dot{1}}$ | 2 $\underline{\dot{1}\ \dot{2}}$ $\underline{\dot{2}}$ 5 | 3 - 0 3
 | 6 - 5 - | $\underline{3\ 2}$ $\dot{1}$ - 0 $\underline{\dot{1}}$ | 2 $\underline{\dot{1}\ \dot{2}}$ $\underline{\dot{2}}$ 7 | 6 - 0 $\underline{6\ 7}$: $\parallel$ 6 - - - |

Fine

图 1. 《天空之城》 数字简谱

```

scale = 2; baseLen = 0.2; fs = 8192;
% tone, noctave, rising, len(eighth)
noteTable = [
    % [0,0,0,6];
    [6,0,0,1];
    [7,0,0,1];

    [1,1,0,3];
    [7,0,0,1];
    [1,1,0,2];
    [3,1,0,2];

    [7,0,0,6];
    [3,0,0,1];
    [3,0,0,1];

    [6,0,0,3];
    [5,0,0,1];
    [6,0,0,2];
    [1,1,0,2];

    [5,0,0,4];
    [0,0,0,2];
  
```

```

    [3,0,0,1];
    [3,0,0,1]
];
music_exp = [];
for m = 1:length(noteTable)
    note = num2cell(noteTable(m, :));
    [tone, nOctave, rising, mulLength] = note{:};
    music_exp = [music_exp, gen_wave_3(tone, scale, nOctave, rising,
mulLength*baseLen, fs)];
end
sound(music_exp, fs)
% audiowrite('output\practice3.wav', music, fs)

```

Practice 4

```

scale = 2; baseLen = 0.2; fs = 8192;
% tone, nOctave, rising, len(eighth)
noteTable = [
    % [0,0,0,6];
    [6,0,0,1];
    [7,0,0,1];

    [1,1,0,3];
    [7,0,0,1];
    [1,1,0,2];
    [3,1,0,2];

    [7,0,0,6];
    [3,0,0,1];
    [3,0,0,1];

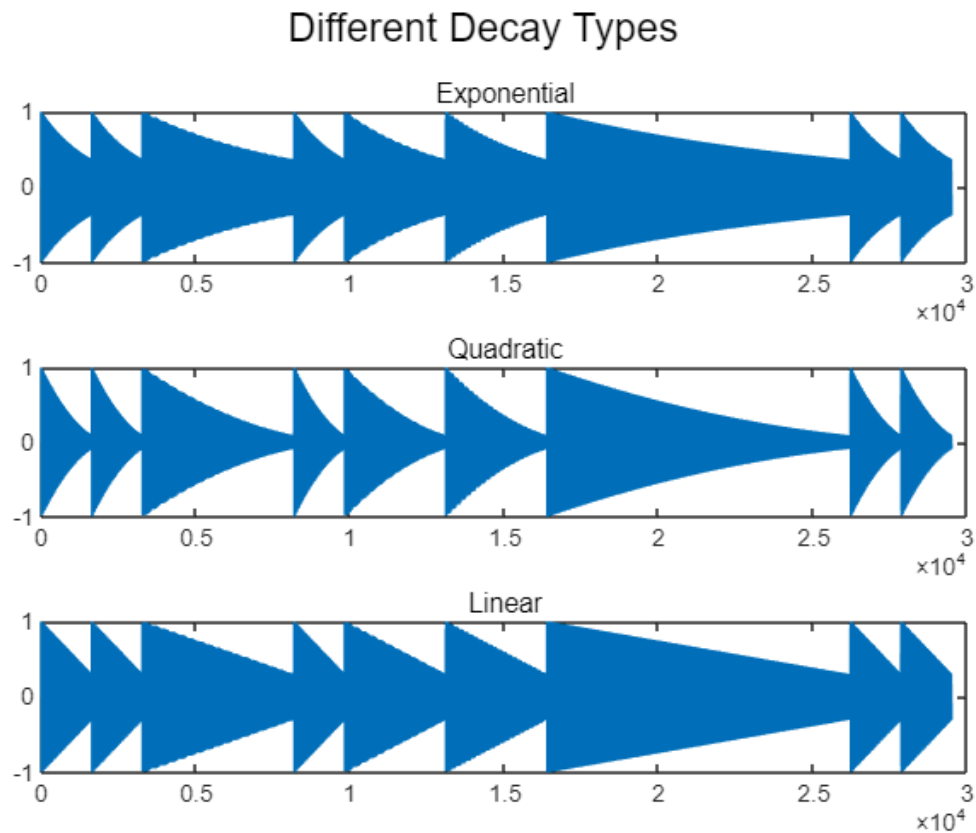
    % [6,0,0,3];
    % [5,0,0,1];
    % [6,0,0,2];
    % [1,1,0,2];
    %
    % [5,0,0,4];
    % [0,0,0,2];
    % [3,0,0,1];
    % [3,0,0,1]
];
music_exp = []; music_linear = []; music_square = [];
for m = 1:length(noteTable)
    note = num2cell(noteTable(m, :));
    [tone, nOctave, rising, mulLength] = note{:};
    music_exp = [music_exp, gen_wave_4(tone, scale, nOctave, rising,
mulLength*baseLen, fs, 'exp')];
    music_square = [music_square, gen_wave_4(tone, scale, nOctave, rising,
mulLength*baseLen, fs, 'square')];
    music_linear = [music_linear, gen_wave_4(tone, scale, nOctave, rising,
mulLength*baseLen, fs, 'linear')];
end
% sound(music_exp, fs)

```

```

subplot(311),sgtitle('Different Decay Types')
plot(music_exp), title('Exponential')
subplot(312)
plot(music_square), title('Quadratic')
subplot(313)
plot(music_linear), title('Linear')

```



```

% audiowrite('output\practice4_exp.wav', music_exp, fs)
% audiowrite('output\practice4_linear.wav', music_linear, fs)
% audiowrite('output\practice4_square.wav', music_square, fs)

```

I prefer linear decay.

```

clear
fs = 8192

```

```

fs = 8192

```

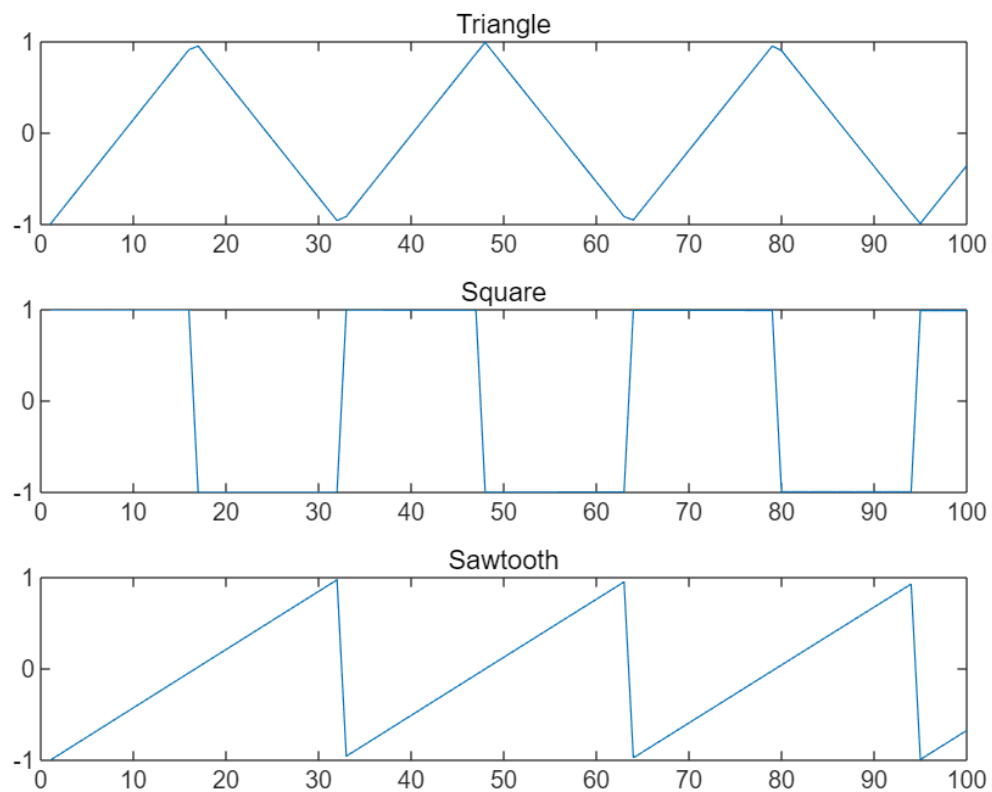
```

% waves = gen_wave_5(tone, scale, noctave, rising, rhythm, fs, decayType,
% harmonyType)
note0 = gen_wave_5(1,1,0,0,1,fs,"linear","triangle");
note1 = gen_wave_5(1,1,0,0,1,fs,"linear","square");
note2 = gen_wave_5(1,1,0,0,1,fs,"linear","sawtooth");
note3 = gen_wave_5(1,1,0,0,1,fs,"linear","piano");
note4 = gen_wave_5(1,1,0,0,1,fs,"linear","flute");
note5 = gen_wave_5(1,1,0,0,1,fs,"linear","comp");

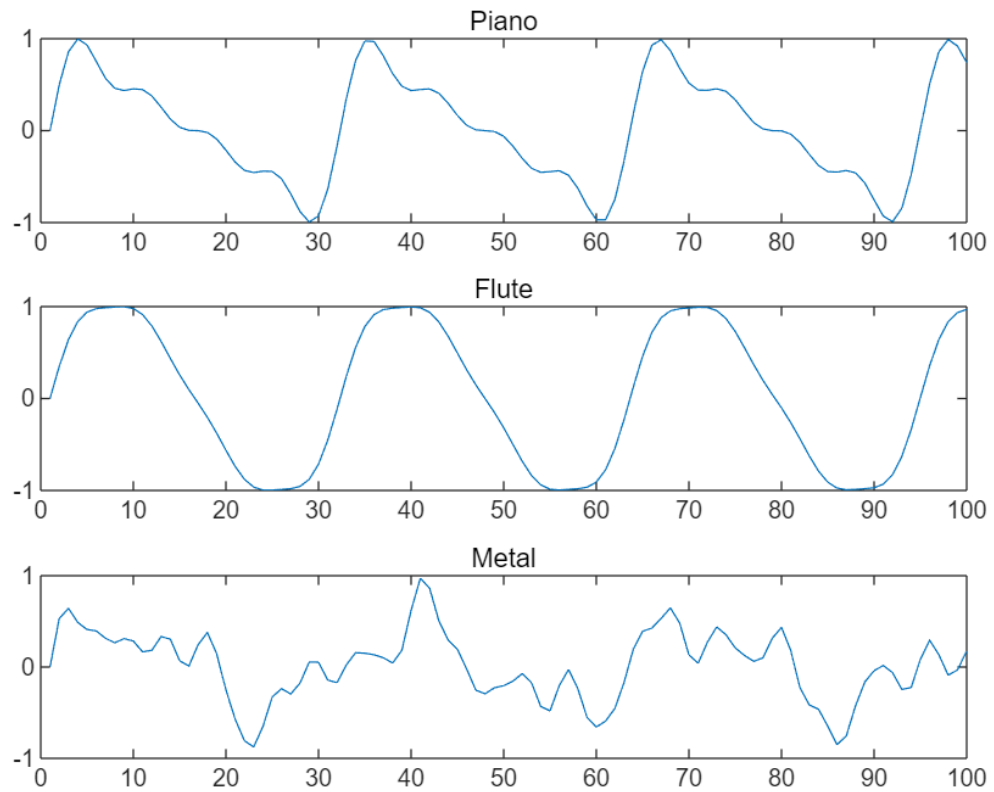
```

```
sound([note3,note4,note5],fs)
% sound(note7,fs)
```

```
figure
subplot(311)
plot(note0(1:100)), title('Triangle')
subplot(312)
plot(note1(1:100)), title('Square')
subplot(313)
plot(note2(1:100)), title('Sawtooth')
```



```
figure
subplot(311)
plot(note3(1:100)), title('Piano')
subplot(312)
plot(note4(1:100)), title('Flute')
subplot(313)
plot(note5(1:100)), title('Metal')
```



```

scale = 1; baseLen = 0.22; fs = 8192;
% tone, noOctave, rising, len(eighth)
noteTable1 = [
    % 1
    [6,0,0,1];
    [6,0,0,1];
    [0,0,0,2];
    [4,0,1,1];
    [4,0,1,1];
    [0,0,0,2];
    % 2
    [0,0,0,3];
    [6,0,0,1];
    [4,0,1,1];
    [3,0,0,1];
    [2,0,1,1];
    [1,0,1,1];
    % 3
    [3,0,0,1];
    [3,0,0,1];
    [0,0,0,2];
    [4,0,1,1];
    [4,0,1,1];
    [0,0,0,2];
    % 4
    [0,0,0,8];
    % 5
    [6,0,0,1];
    [6,0,0,1];
    [0,0,0,2];

```

```
[4,0,1,1];
[4,0,1,1];
[0,0,0,2];
% 6
[0,0,0,3];
[1,0,1,1];
[4,0,1,1];
[3,0,0,1];
[2,0,1,.2];
[3,0,0,.2];
[2,0,1,.6];
[1,0,1,1];
% 7
[3,0,0,1];
[3,0,0,1];
[0,0,0,2];
[4,0,1,1];
[4,0,1,1];
[0,0,0,2];
% 8
[1,0,1,1];
[1,0,1,1];
[1,0,1,1];
[1,0,1,1];
[3,0,0,1];
[3,0,0,1];
[3,0,1,1];
[3,0,1,1];

%9
[1,0,1,2];
[1,0,1,1];
[1,0,1,1];
[1,0,1,2];
[1,0,1,1];
[1,0,1,1];
%10
[1,0,1,1];
[1,0,1,1];
[1,0,1,2];
[6,0,0,2];
[1,0,1,1];
[1,0,1,1];
%11
[3,0,0,1];
[3,0,0,1];
[0,0,0,2]
[3,0,0,1];
[4,0,1,1];
[0,0,0,1];
[1,0,1,5];
%12
[0,0,0,4];

%13
[0,0,0,4]
```


[4,0,1,2]

[4,0,1,1]

[4,0,1,1]

%14

[1,0,1,4]

[0,0,0,4]

%15

[0,0,0,4]

[7,-1,0,1]

[7,-1,0,1]

[1,0,1,2]

%16

[7,-1,0,4]

[0,0,0,4]

%17

[1,0,1,2];

[1,0,1,1];

[1,0,1,1];

[1,0,1,2];

[1,0,1,1];

[1,0,1,1];

%18

[1,0,1,1];

[1,0,1,1];

[1,0,1,2];

[6,0,0,2];

[1,0,1,1];

[1,0,1,1];

%19

[3,0,0,1]

[3,0,0,1]

[3,0,0,1]

[1,0,1,1]

[3,0,0,2]

[4,0,1,1]

[1,0,1,3]

%20

[0,0,0,6]

%21

[0,0,0,4]

[4,0,1,1]

[4,0,1,1]

[4,0,1,2]

%22

[4,0,1,4]

[0,0,0,2]

[5,0,1,1]

[6,0,0,1]

%23

[5,0,1,1]

[5,0,1,1]

[6,0,0,2]

[5,0,1,2]

[0,0,0,2]

%24

```

[0,0,0,4]
[6,0,0,2]
[5,0,1,2]
%END
[1,0,1,1]
[2,0,0,1]
[3,0,0,1]
[5,0,1,4]

];

noteTable2 = [
    % 1
    [4,-1,1,1];
    [4,-1,1,1];
    [0,0,0,2];
    [4,-2,1,1];
    [4,-2,1,1];
    [0,0,0,2];
    %2
    [0,0,0,8];
    %3
    [1,-1,1,1];
    [1,-1,1,1];
    [0,0,0,2]
    [1,-1,1,1];
    [1,-1,1,1];
    [0,0,0,2];
    %4
    [1,-1,1,1];
    [1,-1,1,1];
    [1,-1,1,1];
    [1,-1,1,1];
    [3,-1,0,1];
    [3,-1,0,1];
    [3,-1,1,1];
    [3,-1,1,1];
    %5
    [4,-1,1,1];
    [4,-1,1,1];
    [0,0,0,2];
    [4,-2,1,1];
    [4,-2,1,1];
    [0,0,0,2];
    %6
    [0,0,0,8];
    %7
    [1,-1,1,1];
    [1,-1,1,1];
    [0,0,0,2]
    [1,-1,1,1];
    [1,-1,1,1];
    [0,0,0,2];
    %8
    [1,-1,1,1];
    [1,-1,1,1];

```

```
[1,-1,1,1];  
[1,-1,1,1];  
[3,-1,0,1];  
[3,-1,0,1];  
[3,-1,1,1];  
[3,-1,1,1];
```

%9

```
[4,-2,1,1]  
[1,-1,1,1]  
[4,-1,1,2]  
[4,-2,1,1]  
[1,-1,1,1]  
[4,-1,1,2]
```

%10

```
[4,-2,1,1]  
[1,-1,1,1]  
[4,-1,1,2]  
[4,-2,1,1]  
[1,-1,1,1]  
[4,-1,1,2]
```

%11

```
[1,-1,1,1]  
[3,-1,0,1]  
[5,-1,1,1]  
[3,-1,0,1]  
[1,-1,1,1]  
[3,-1,0,1]  
[5,-1,1,1]  
[3,-1,0,1]
```

%12

```
[1,-1,1,1]  
[3,-1,0,1]  
[5,-1,1,1]  
[3,-1,0,1]  
[1,-1,1,1]  
[3,-1,0,1]  
[3,-1,0,1]  
[3,-1,1,1]
```

%13

```
[4,-1,1,1]  
[6,-1,0,1]  
[1,0,1,1]  
[6,-1,0,1]  
[4,-1,1,1]  
[6,-1,0,1]  
[1,0,1,2]
```

%14->22

```
[4,-2,1,1]  
[1,-1,1,1]  
[4,-1,1,1]  
[1,-1,1,1]  
[4,-2,1,1]  
[1,-1,1,1]
```

[4,-1,1,1]

[1,-1,1,1]

%15

[3,-2,0,1]

[7,-2,0,1]

[3,-1,0,1]

[7,-2,0,1]

[3,-2,0,1]

[7,-2,0,1]

[3,-1,0,2]

%16

[3,-2,0,1]

[7,-2,0,1]

[3,-1,0,1]

[5,-1,1,1]

[1,-1,1,1]

[3,-1,0,1]

[5,-1,1,1]

[1,0,1,1]

%17

[4,-1,1,1]

[6,-1,0,1]

[1,0,1,2]

[4,-1,1,1]

[6,-1,0,1]

[1,0,1,2]

%18

[4,-1,1,1]

[6,-1,0,1]

[1,0,1,2]

[4,-1,1,1]

[6,-1,0,1]

[1,0,1,2]

%19->27

[1,-1,1,1]

[3,-1,0,1]

[5,-1,1,2]

[1,-1,1,1]

[3,-1,0,1]

[5,-1,1,2]

%20

[1,-1,1,1]

[3,-1,0,1]

[5,-1,1,1]

[3,-1,0,1]

[1,-1,1,1]

[3,-1,0,1]

[5,-1,1,1]

[3,-1,0,1]

%21

[4,-2,1,1]

[1,-1,1,1]

[4,-1,1,1]

[1,-1,1,1]

[4,-2,1,1]

[1,-1,1,1]

```

[4,-1,1,2]
%22
[4,-2,1,1]
[1,-1,1,1]
[4,-1,1,1]
[1,-1,1,1]
[4,-2,1,1]
[1,-1,1,1]
[4,-1,1,1]
[1,-1,1,1]
%23
[3,-2,0,1]
[7,-2,0,1]
[3,-1,0,2]
[3,-2,0,1]
[7,-2,0,1]
[3,-1,0,1]
[7,-2,0,1]
%END
[3,-2,0,1]
[5,-2,1,1]
[7,-2,0,1]
[3,-1,0,1]
[4,-1,1,2]
[1,-1,1,2]

];
music1 = [];
for m = 1:length(noteTable1)
    note = num2cell(noteTable1(m, :));
    [tone, noctave, rising, mulLength] = note{:};
    music1 = [music1, ...
        gen_wave_5(tone, scale, noctave, rising, mulLength*baseLen, fs, ...
            'square', 'triangle')];
end
% music1

music2 = [];
for m = 1:length(noteTable2)
    note = num2cell(noteTable2(m, :));
    [tone, noctave, rising, mulLength] = note{:};
    music2 = [music2, ...
        gen_wave_5(tone, scale, noctave, rising, mulLength*baseLen, fs, ...
            'square', 'square')];
end
music2 = music2 * .3

```

```
music2 = 1x345986
    0.3000    0.2998    0.2995    0.2993    0.2991    0.2988    0.2986    0.2984
    0.2981    0.2979    0.2977    0.2974    0.2972    0.2970    0.2967    0.2965
0.2963    0.2960    0.2958    0.2956    0.2954    0.2951    0.2949   -0.2947
-0.2944   -0.2942   -0.2940   -0.2937   -0.2935   -0.2933   -0.2930   -0.2928
-0.2926   -0.2924   -0.2921   -0.2919   -0.2917   -0.2914   -0.2912   -0.2910
-0.2907   -0.2905   -0.2903   -0.2901   -0.2898    0.2896    0.2894    0.2891
0.2889    0.2887
```

```
N1 = length(music1)
```

```
N1 = 358600
```

```
N2 = length(music2)
```

```
N2 = 345986
```

```
if N1 > N2
    music2 = [music2,zeros(1,N1-N2)]
elseif N1 < N2
    music1 = [music1,zeros(1,N2-N1)]
end
```

```
music2 = 1x358600
    0.3000    0.2998    0.2995    0.2993    0.2991    0.2988    0.2986    0.2984
    0.2981    0.2979    0.2977    0.2974    0.2972    0.2970    0.2967    0.2965
0.2963    0.2960    0.2958    0.2956    0.2954    0.2951    0.2949   -0.2947
-0.2944   -0.2942   -0.2940   -0.2937   -0.2935   -0.2933   -0.2930   -0.2928
-0.2926   -0.2924   -0.2921   -0.2919   -0.2917   -0.2914   -0.2912   -0.2910
-0.2907   -0.2905   -0.2903   -0.2901   -0.2898    0.2896    0.2894    0.2891
0.2889    0.2887
```

```
% sound(music1, fs)
% sound(music2, fs)
sound(music1+music2, fs)
```

```
audiowrite('output\Грынна_крови.wav', music1+music2, fs)
```

警告：数据在写入文件期间被裁剪。

